

Reuben Docea

CV

Edinburgh, Scotland

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[Google Scholar](#)



Throughout my career, I have pursued work that combines promising technology with the possibility of tangible impact. This has led me from founding a university society to 3D-print prosthetic devices for children in Glasgow, through agricultural robotics, to six years of research at the interface of computer vision and minimally invasive surgery. I now lead technology at Applied Computer Vision, developing and deploying computer vision systems for industrial settings. The role has broadened my work beyond engineering into client conversations and practical delivery. I am most interested in work that combines technical depth with a clear purpose, particularly where computer vision, robotics and AI can improve how people understand and act on the physical world.

Education

2014–2019 **MEng Biomedical Engineering, 1st Class**, [The University of Glasgow](#), Scotland.

Employment Record

2025–Present **Head of Technology (CV & ML Lead)**, [Applied Computer Vision](#), Edinburgh, Scotland.
Early technical hire leading applied computer vision and ML engineering alongside client-facing work. Developed and deployed industrial vision systems, established reusable engineering and delivery practices, and represented ACV externally, including at Campden BRI.

2020–2026 **PhD (submitted)**, [National Center for Tumor Diseases](#), Dresden, Germany.

Thesis: "*Computer Assisted Navigation for Laparoscopic Surgery in the Context of Soft Tissue Deformation*"

AI-driven computer vision methods for surgical navigation systems (publications - [Google Scholar](#)).

- **ARAILIS**: Led the ROS-based integration of a navigation system for minimally invasive liver surgery ([video demonstration](#)), focusing on SLAM. This work produced six research publications, a ROSCon 2022 keynote, and a [public code repository](#).
- **Surgical Data Collection**: Initiated and led a project with the Chinese University of Hong Kong and University Hospital Dresden to gather an in-vivo dataset for 4D reconstruction evaluation. I built a complex ROS2 system to synchronize a da Vinci Xi Endoscope, a Structured Light Camera, and Optical Tracking, handling all stereo and hand-eye calibrations. I oversaw time-sensitive data collection in the operating room, working closely with surgical staff to ensure the process was smooth and effective, and often communicating in both English and German.
- **CloudSkin**: Developed intelligent orchestration infrastructure for an EU project with Dell; integrated AI surgery methods into a streaming platform and built a Kubernetes+Docker cluster simulating a multi-operating room environment.
- **NavigationBox**: Co-developed an AR-based surgical navigation solution using ROS2 and AI.
- **Student Supervision**: Supervised multiple Master's/Bachelor's theses and group projects. Two years of coordinating Group Projects modules ([Komplexxpraktikum & Team Project](#)).

2019–2020 **Robotics Engineer**, *Small Robot Company*, Salisbury, England.

Engineered and integrated advanced vision systems to expand robot capabilities, working in close collaboration with computer vision and data science experts.

- Developed C++ & ROS applications for imaging hardware, including stereo-calibration and matching with custom stereo cameras.
- Led camera/lens specification and conducted hyperspectral imaging experiments to inform automated weeding solutions.

Experience & Achievements

- 2018-2019 **Masters Project: Signal Processing and Statistics**, *Biomechatronics and Neurorehabilitation Laboratory of Chalmers University of Technology*, Gothenburg, Sweden.
Independently researched, planned and carried out 6 month project to completion for improving real-time Machine-Learning-based control of motorised prostheses.
 - MATLAB for statistics, signals processing and pattern recognition
- 2017-2018 **Founder and President of [Handprints - e-NABLE Scotland](#)**, *The University of Glasgow*, Glasgow, Scotland.
As founder and president, I established and led a student society that designed and 3D-printed prosthetic devices for local individuals and ran educational initiatives in schools. It still receives [enthusiasm and support](#) from academics and students to this day.
 - Secured a *Chancellor's Fund* Grant for the purchase of a 3D Printer.
 - Won the *Let's Make It Happen* video competition grant.
 - Nominated for a national *Herald Higher Education* award.
 - Led and coordinated multiple student groups through the prosthetic design process.
 - Designed a novel 3D-printed prosthetic device for cycling.

Technical Skills

Languages	C++, MATLAB, Python
Computer Vision & ML	Detection, segmentation, tracking, SLAM; stereo & hand-eye calibration; edge deployment
Frameworks	AWS (RDS, Lambda, Elastic Beanstalk), CVAT, Docker, Git/GitHub, Kubernetes, Linux, ROS1/ROS2
Software	Solidworks, Meshmixer, LaTeX

Interests - some of many

Cycling	Trips with friends: Austria & Czechia, Switzerland, and the Black Forest
Board Games	I enjoy board games and group social deduction games, and organise a regular board games evening at work.
Creative Pursuits	3D Modeling (partly), 3D printing & painting a 'trophy' for Kemet ; Building an outdoor Fire-court from stone.

Languages

English	Native
Romanian	Native
German	B2

References

Academic	Past Employer
◦ Dr. Julien Reboud ↗ ◦ Prof. Dr. Stefanie Speidel ↗	◦ Thomas Burrell ↗